

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Lecture 1: Scope of the Standard Theory

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Overview

We define the preferred F-01 foliation, fix the spatial metric, and recover the Poisson equation in the weak-field limit.

1 Physical assumptions

All coordinate conditions and boundary conditions are stated before a parameter is fitted.

$$g_{ij} = \delta_{ij}$$

2 Field equations

The first two displayed relations establish the notation used in the derivation.

$$\nabla^2 \Phi = 4\pi G \rho$$

3 Observable limit

Weak-field, static, and point-source limits are checked independently.

$$R_a = (O_a - T_a(\theta))/\sigma_a$$

4 Parameter dependence

Parameter combinations are compared through derivatives of the reference observables.

5 Example

The calculation is repeated for one analytic case and one tabulated data point.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Hall, ch. 1
- SGT9, pp. 1–18
- Einstein–Mythos paper E, §§1–2